

Lesson 1 — Meet the Kit & First Press

Matches LEGO Teacher Portal lesson: "Mornings in Motion".

Objective (I can...): I can find my way around the Coding Canvas with pictures, work with a partner, and press two picture blocks together to make a motor move. **CS concept:** What a program is (steps a machine follows) · **Builds on:** No prior lesson — course launch.

AI Connection (2 min, read aloud): A tablet that unlocks when it sees your face is a computer following steps and making a choice. *Ask the group:* "If a robot wanted to know 'is this a cat?', what tiny things could it look at?" (Just collect ideas — we'll come back to this all year.)

Materials: LEGO CS & AI kit 45520 (1 per pair), a device per pair (Coding Canvas, PIN ready — no login), a big printed "kit parts" picture card per pair, a "team jobs" card (Driver / Navigator pictures). **No full build today** — use the kit's quick-start model (hub plus one motor already on). The unit's real model is built next lesson.

60-minute flow - Warm-up (10): "Robot chef" — a child gives the coach spoken steps to make a pretend sandwich; the coach does exactly what's said, silly mistakes and all (butters the plate, etc.). Big point, lots of giggles: machines do EXACTLY what you tell them. Then teach the two team jobs with a quick chant: "Driver drives, Navigator points." - **Concept + demo (8):** Turn on the hub; show it connecting to the tablet. Point to the picture palette, the big green Run, and the red Stop. Drag a start picture + one "spin motor" picture; press Run so a real motor moves. Let them gasp. - **Build/Code (25):** Each pair (1) gets their hub connected with coach help, (2) takes a picture tour — find the palette, Run, Stop, (3) places 2 picture blocks (start + spin) on the quick-start model and presses Run. *Wiggle break + swap at ~12 min, and practice trading the two jobs.* Coach circulates so every hub is connected. - **Challenge (7):** Add a "stop after a moment" picture block so the motor stops on its own. Change which picture is first and guess what happens. - **Share + tidy-up (10):** Each pair presses Run for the group and points to one kit part by its picture. Coil cables, hub off, parts in the tray. Sing a "clean-up" count-down.

Pair roles: Driver connects (with help) and drags pictures; Navigator holds the parts card and finds each part. Swap at the wiggle break. *(Trio: add a Tester who presses Run and tells the group what moved.)*

Differentiation: *Support* — a palette card so kids match blocks by picture, not words. *Extension / fast-finisher* — add a sound picture when the motor stops. *Reteach* — connect the hub together slowly; have the child name each button before pressing.

Exit check: Pair has a working hub, ran a 2-block program that moves a motor, can point to the hub, a motor, and the green Run, and can name both team jobs. **At-home (optional):** Tell a grown-up 4 exact steps to put on your shoes, like talking to a robot. ~5 min.